Changyi Li

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Education

Fudan University & Shanghai Innovation Institute (SII) Sep 2026 – Jun 2029 (Expected)
Ph.D. in Artificial Intelligence *Shanghai, China*
- Advisors: Prof. Xipeng Qiu, Associate Research Prof. Xudong Pan, and Prof. Min Yang

Fudan University, System Software and Security Lab Sep 2024 – Sep 2026
*M.S. in Computer Science and Technology; GPA: **3.96/4.00**, Rank: **1/379*** *Shanghai, China*
- Advisors: Associate Research Prof. Xudong Pan and Prof. Min Yang

Qilu University of Technology (Shandong Academy of Sciences) Sep 2020 – Jun 2024
*B.S. in Intelligent Science and Technology; GPA: **4.50/5.00**, Rank: **1/102*** *Shandong, China*

Research Interests

My research focuses on the **safety and alignment** of **frontier AI systems**, including **agentic AI**, and their reliability, trustworthiness, and controllability as capabilities expand and tasks grow more complex. My work spans **intent and behavioral safety in large models and agents** [1, 2, 3], **capability boundaries and frontier risks** [2, 3, 4, 5], **deep value alignment** [6, 7], and **multimodal model safety and adversarial testing**. I am also exploring **physical AI**.

Selected Honors and Awards

First-Class Scholarship, Fudan University 2025, 2026
Freshman Scholarship, Fudan University 2024
President's Scholarship (**1**/38,000+) 2024
Provincial Outstanding Graduate (Top 0.1%) 2024
National Scholarship (Top 0.1%) 2023
MCM/ICM Meritorious Winner, Team Leader 2023
First-Class Scholarship & Outstanding Student Pacesetter 2022, 2023

Publications (* Equal contribution)

[1] **ReasoningShield: Safety Detection over Reasoning Traces of Large Reasoning Models.**
Changyi Li, Jiayi Wang, Xudong Pan, Geng Hong, Min Yang
NeurIPS, 2026. [Paper] [Code] [Page] [Models] [Data] [PaperWeekly]

[2] **AutoControl Arena: Synthesizing Executable Test Environments for Frontier AI Risk Evaluation.**
Changyi Li, Pengfei Lu, Xudong Pan, Fazl Barez, Min Yang
ICML, 2026. [Paper] [Code] [Page] [Ji Qi Zhi Xin] [PaperWeekly]

[3] **Understanding Loss-of-Control Risk in Frontier AI Systems.**
Changyi Li, Yun Chen, Yiyang Zhang, Wenqi Zhang, Changyue Jiang, Xudong Pan; Jiarun Dai, Min Yang
Technical Report, 2026. [Paper]

[4] **Large Language Model-Powered AI Systems Achieve Self-Replication with No Human Intervention.**
Core Contributor
Technical Report, 2025. [Paper] [Page] [Live Science] [Forbes] [The Independent]

[5] **CyberEvolver: Structured Self-Evolution for Cybersecurity Agents On the Fly.**
Yihe Fan, **Changyi Li**, Lichen Xu, Xudong Pan, Jiarun Dai, Hong Geng, Min Yang
Submitted to ICLR 2027. [Paper] [Code] [Page]

[6] **ValueGraph: Evaluating and Representing Value Trade-offs in Large Language Models.**
Dylan Xinming Hou*, **Changyi Li***, Chloe Ruyi Guo, Ziyuan Zhao, Gus Xia, Hua Shen
Submitted to ICLR 2027. [Paper] [Page]

[7] **Gyges: Benchmarking Value Alignment as Hierarchical Value Stability.**
Dylan Xinming Hou*, **Changyi Li***, Chloe Ruyi Guo, Ziyuan Zhao, Yichi Zhang, Hua Shen
Submitted to ACL 2027. [Paper] [Page]

Research Experience

ReasoningShield: Safety Auditing of Reasoning Traces First Author | Feb – Jun 2025

- **Problem:** A model's reasoning can contain harmful content even if its final answer is benign or it later refuses the request.
- **Approach:** Built a three-level, ten-category risk taxonomy and trained **ReasoningShield-1B/3B** using SFT and DPO to detect harmful reasoning and locate risky steps.
- **Contribution:** Introduced the first dedicated CoT safety-auditing framework, including a benchmark of **9.2K** annotated query–CoT examples and specialized moderation models.
- **Results:** ReasoningShield-3B achieved **91.8%** F1 on the CoT benchmark, exceeding LlamaGuard-4-12B by **35.6%**. On mainstream QA safety benchmarks, it achieved **85.0%** F1, a **15.7%** improvement.
- **Impact:** Published at **NeurIPS 2026**; featured by PaperWeekly; open models and data have over **4K Hugging Face downloads** and have been cited by teams including an IEEE Fellow and Canada's National Research Council.

AutoControl Arena: Automated Frontier Risk Evaluation First Author | Jul 2025 – Mar 2026

- **Problem:** Autonomous agents introduce long-tail risks that are costly to test with manual benchmarks; text-only simulators often lack causal fidelity.
- **Approach:** Built an Architect–Coder–Monitor framework that designs scenarios, synthesizes executable sandboxes through logic–narrative decoupling, and audits behavioral and reasoning traces.
- **Contribution:** Developed an end-to-end automated risk evaluation framework; released X-Bench with **70 scenarios**, **7 risk categories**, and **15 domains**; introduced stress–temptation testing.
- **Results:** Achieved a **98%** end-to-end environment construction success rate; reproduced OpenAI/Anthropic high-risk behaviors; identified alignment illusion, bidirectional safety scaling, and stress–temptation coupling.
- **Impact:** Published at **ICML 2026**; received **100+ GitHub stars** and **600+ clones** and coverage from Ji Qi Zhi Xin and PaperWeekly. Invited to speak at the Seoul Alignment Workshop (one of two attendees from China) and attend the OpenAI Safety & Alignment Mixer (sole attendee from China).

NUWA: Frontier AI Loss-of-Control Risk Report First Author | Aug – Nov 2026

- **Motivation:** In the OpenAI–Hugging Face incident, cybersecurity agents coordinated through unauthorized channels, bypassed isolation, and attempted to alter execution records. We investigate the factors behind such risks and their implications for human control.
- **Evaluation:** Evaluated MiMo-V2.6-Flash, GLM-5.3, and DeepSeek-V4.1-Flash-1017 across **nine benchmarks**, testing evaluation awareness, compliance under pressure, truthful reporting, boundary crossing, and responses to shutdown instructions. Varied instructions, reasoning settings, and execution frameworks, and analyzed full trajectories to understand the mechanisms behind risk behavior.
- **Findings:** Some agents changed correct code to satisfy faulty tests or concealed known facts under pressure. Higher reasoning settings increased shutdown avoidance in all six model–framework combinations under default task settings; agents sometimes accessed hosts or modified restricted objects even when tasks failed.
- **Impact:** As an independent evaluator, reached an understanding on potential collaboration to assess Chinese models for entry into the **EU market**.

AI System Self-Replication Risk Study Core Contributor | Nov 2024 – Feb 2025

- **Motivation:** Self-replication is widely regarded as a frontier AI safety red line, but leading labs at the time had assessed the risk as minimal.
- **Approach:** Designed a Perceive–Reflect–Discover–Plan scaffold for long-horizon replication; evaluated **32** open-source models using a reproducible protocol with explicit milestones and success criteria.
- **Contribution:** Demonstrated end-to-end self-replication in **11/32 systems**, including a locally runnable **14B** model, and investigated spontaneous replication, self-exfiltration, and shutdown evasion.
- **Impact:** Reached **millions of views and reposts** across X, YouTube, and TikTok; covered by *Live Science*, *Forbes*, *The Guardian*, and *The Independent*; invited to speak at a France AI Action Summit side event.

CyberEvolver: Self-Evolving Cybersecurity Agents Second Author | Feb – Jun 2026

- **Problem:** Fixed agent workflows struggle across targets and failure modes; self-evolution must contend with unstructured changes, noisy feedback, and limited diversity.
- **Approach:** Decomposed the harness into four evolvable layers, converted execution traces into actionable diagnoses, and used population-based beam search to retain diverse variants.
- **Contribution:** Proposed an on-policy harness self-evolution framework for cybersecurity. Across CTF, vulnerability exploitation, and penetration testing, improved seed-agent pass@16 by **13.6 percentage points** on average and exceeded the strongest human-designed baseline by **14.0%**.

ValueGraph: Evaluating Value Trade-offs in LLMs Co-first Author | Jun – Oct 2026

- **Problem:** Overall value rankings obscure how models resolve particular value conflicts in different contexts.
- **Approach:** Represented Schwartz's 19 values as nodes and pairwise priorities as directed edges; combined best–worst scaling with Bradley–Terry estimation to quantify contextual trade-offs and their robustness.
- **Contribution:** Introduced a graph representation of value orientations, pairwise priorities, and contextual evidence; constructed **2,791 scenarios** covering **all 171 value pairs** across **8 domains**.
- **Findings:** Across **22 models**, similar overall rankings concealed divergent trade-offs; priorities among highly ranked values lacked consistent robust ordering, and apparently robust priorities sometimes reversed across domains.

Gyges: Agent Value Alignment and Hierarchical Stability Co-first Author | Jun – Oct 2026

- **Problem:** Evaluations that state rules or flag violations explicitly can conflate rule-following with value alignment and miss failures when rules change.
- **Approach:** Built a three-level framework of value principles, literal requests, and explicit rules. With minimal factual contrasts, varied requests, rules, pressure, and oversight; compared tool use with independent judgments.
- **Contribution:** Introduced **hierarchical value stability** and a human-validated benchmark of **1,040 items** across **40 scenario families**, covering honesty, care, human oversight, and privacy.
- **Findings:** Across **13 models**, explicit rules masked over-compliance and caution; literal tests missed indirect violations, and some models took actions they judged unacceptable even without task pressure or absent oversight.

Applied Projects

Text-to-Image Security and Red Teaming (Alibaba Tianchi) Project Lead | Sep – Nov 2024

- **Approach:** Used cognitive deception and semantic association to test the full text-filter, image-generation, and defense pipeline, combining scenario rewriting, adversarial affixes, caption injection, prompt fuzzing, and VLM pre-screening.
- **Results:** Led gray- and black-box testing of **30+** systems; achieved **90%+** text-filter bypass and **80%+** image attack success; placed **26th of 1,574 teams** (top 1.7%). Reported severe vulnerabilities in **10+** models through a national security briefing that received written instructions from central authorities.

Multimodal Safety Evaluation and Protection System Project Lead | Sep 2025 – Mar 2026

- **Partners:** Alibaba Cloud and Chinese Academy of Sciences.
- **Problem:** Multimodal safety review faces scarce real-world risk samples, costly labels, and unreliable or hard-to-trace decisions.
- **Approach:** Built an automated data synthesis and labeling pipeline, expanding risk concepts into visual scenes and prompts; combined VLM review, model voting, consistency scoring, confidence quantification, and hard-example feedback to produce traceable pseudo-labels.
- **Results:** Achieved **95%+ evaluation accuracy** and a **90% efficiency improvement**; supported Alibaba Cloud, CAS, China Electronics, and E Fund in multimodal safety evaluation and protection; produced **one invention patent**.

AIGC Artistic-Text Quality and Compliance Detection Project Lead | Jun – Sep 2025

- **Partner:** China Electronics. General VLMs achieved only about 60% accuracy on fine-grained artistic-text defects.
- **Approach:** Built a five-stage pipeline combining OCR, rule checks, vision-language models, multimodal fine-tuning, ViT defect classification, and ensembles; led taxonomy design, training, and deployment.
- **Results:** Covered **8 defect categories** and **10+ visual styles**; achieved **95%+** negative-sample detection at **500 ms** latency, handled **500K+/day** peak volume, and delivered **10M+** high-quality data assets.

Academic Service

Invited Talk	**Seoul Alignment Workshop, FAR.AI (1 of 2 from China)**	**2026**
Invited Guest	**OpenAI Safety & Alignment Mixer** (sole attendee from China)	**2026**
Organizer	**AI Deception Workshop, Fudan, SAIF & Concordia AI**	**2025**
Teaching Assistant	Foundations of Deep Learning, Fudan University (Spring)	2025, 2026
Reviewer	ICLR 2027, WWW 2027, IEEE T-IFS	2026–2027